

100 Space Terms and Definitions

A collection of 100 important space and astronomy terms with detailed definitions for study, revision, and academic reference.

1. Universe

The universe is the totality of space, time, matter, energy, galaxies, and the physical processes that govern them. Modern observations show that the universe is expanding and contains enormous structures separated by vast distances. Scientists study its origin, evolution, composition, and possible future through cosmology and astronomy.

2. Cosmos

The cosmos is a broad term used to describe the universe as an ordered and interconnected system. It includes galaxies, stars, planets, cosmic radiation, dark matter, and dark energy. The study of the cosmos combines observations and physical theories to understand how the universe formed and how its large-scale structures developed.

3. Galaxy

A galaxy is a huge gravitationally bound collection of stars, gas, dust, and dark matter. Galaxies occur in different forms, including spiral, elliptical, and irregular shapes. They can contain millions, billions, or even trillions of stars. Galaxies interact with one another and can merge over extremely long periods of time.

4. Milky Way

The Milky Way is the spiral galaxy that contains our Solar System. It contains hundreds of billions of stars along with gas, dust, and a large amount of dark matter. The Solar System lies in one of its spiral arms, far from the galactic center. The Milky Way is part of a larger collection of galaxies called the Local Group.

5. Solar System

The Solar System is the gravitational system centered on the Sun. It includes eight planets, dwarf planets, moons, asteroids, comets, dust, and other smaller bodies. The Sun contains most of the system's mass and provides the energy that drives many processes on Earth. The Solar System formed about 4.6 billion years ago.

6. Sun

The Sun is a star at the center of our Solar System and the main source of energy for Earth. It is composed mostly of hydrogen and helium and produces energy through nuclear fusion in its core. The Sun's light and heat influence Earth's climate and support life. Solar activity also produces phenomena such as solar flares and the solar wind.

7. Star

A star is a massive, luminous object made mainly of extremely hot plasma. Its gravity holds the star together while energy produced in its interior supports it against gravitational collapse. Most stars generate energy through nuclear fusion. Stars vary greatly in mass, temperature, brightness, color, and lifetime.

8. Planet

A planet is a large celestial body that orbits a star and is massive enough for gravity to make it approximately spherical. In our Solar System, planets are divided broadly into rocky inner planets and giant outer planets. Planets can have atmospheres, magnetic fields, rings, and moons depending on their formation and evolution.

9. Dwarf Planet

A dwarf planet is a celestial body that orbits the Sun, is nearly spherical because of its gravity, but has not cleared its orbital neighborhood of other objects. Pluto, Eris, Haumea, Makemake, and Ceres are recognized dwarf planets. They provide valuable information about the formation and evolution of planetary systems.

10. Moon

A moon is a natural satellite that orbits a planet, dwarf planet, or sometimes another small body. Moons form through different processes, including accretion, capture, and giant impacts. They can have diverse surfaces and geological histories. Earth's Moon strongly influences ocean tides and stabilizes some aspects of Earth's rotational motion.

11. Mercury

Mercury is the smallest planet in the Solar System and the closest planet to the Sun. It has a rocky surface covered with impact craters and experiences very large temperature differences between day and night. Mercury has a thin exosphere rather than a substantial atmosphere. Its year lasts only about 88 Earth days.

12. Venus

Venus is the second planet from the Sun and is similar to Earth in size and composition. Its thick atmosphere is dominated by carbon dioxide and produces an intense greenhouse effect. Venus has extremely high surface temperatures and clouds containing sulfuric acid. It rotates slowly and in the opposite direction to most planets.

13. Earth

Earth is the third planet from the Sun and the only known world to support life. It has abundant liquid water, a protective atmosphere, and a magnetic field. Its surface is divided into continents and oceans

and is shaped by geological activity. Earth's atmosphere and climate create conditions suitable for a remarkable diversity of organisms.

14. Mars

Mars is the fourth planet from the Sun and is commonly called the Red Planet because iron minerals on its surface have oxidized. It has a thin carbon dioxide-rich atmosphere, polar ice caps, volcanoes, valleys, and impact craters. Evidence shows that liquid water existed on its surface in the distant past. Mars is a major target for robotic exploration.

15. Jupiter

Jupiter is the largest planet in the Solar System and is a gas giant composed primarily of hydrogen and helium. It has a powerful magnetic field, numerous moons, and a faint ring system. Its atmosphere contains large storms, including the famous Great Red Spot. Jupiter's enormous gravity strongly influences the motion of smaller Solar System bodies.

16. Saturn

Saturn is the sixth planet from the Sun and is famous for its extensive system of bright rings. It is a gas giant composed mainly of hydrogen and helium and has many moons. Saturn is less dense than water on average. Its moon Titan has a thick atmosphere and lakes and seas of liquid hydrocarbons on its surface.

17. Uranus

Uranus is the seventh planet from the Sun and is classified as an ice giant. Its atmosphere contains hydrogen, helium, and methane, with methane contributing to its blue-green appearance. Uranus rotates with an extreme axial tilt, making its seasons unusual. It also has a faint ring system and numerous moons.

18. Neptune

Neptune is the eighth and most distant major planet in the Solar System. It is an ice giant with a deep atmosphere containing hydrogen, helium, and methane. Neptune has extremely fast winds and powerful storms. Its large moon Triton is believed to have been captured from the outer Solar System.

19. Pluto

Pluto is a dwarf planet located in the distant Kuiper Belt beyond Neptune. It has a complex surface containing frozen nitrogen, methane, and water ice. Pluto has mountains, plains, glaciers, and a thin atmosphere that changes with its orbit. NASA's New Horizons spacecraft provided the first close observations of Pluto in 2015.

20. Asteroid

An asteroid is a relatively small rocky or metallic body that orbits the Sun. Most known asteroids are concentrated in the asteroid belt between Mars and Jupiter, although many have different orbital paths. Their sizes and shapes vary greatly. Asteroids preserve material from the early Solar System and are studied to understand planetary formation.

21. Comet

A comet is a small celestial body made primarily of ice, dust, and rocky material. When it approaches the Sun, heating causes some of its ice to turn into gas, producing a coma and often a long tail. Comet tails generally point away from the Sun because of solar radiation and the solar wind. Comets contain ancient material from the early Solar System.

22. Meteoroid

A meteoroid is a relatively small piece of rock or metal traveling through space. Meteoroids can originate from asteroids, comets, or collisions between larger bodies. If one enters Earth's atmosphere and produces a visible streak of light, it is called a meteor. If part of it reaches the ground, the recovered object is called a meteorite.

23. Meteor

A meteor is the bright streak of light produced when a meteoroid enters a planet's atmosphere and heats the surrounding gas. The glowing phenomenon is sometimes called a shooting star, although it is not actually a star. Most small meteoroids burn up completely. Larger objects may survive the passage and produce meteorites.

24. Meteorite

A meteorite is a piece of extraterrestrial material that survives passage through an atmosphere and reaches the surface of a planet or moon. Most meteorites found on Earth originate from asteroids, while a smaller number come from the Moon or Mars. Scientists study meteorites to learn about the early Solar System and planetary materials.

25. Nebula

A nebula is a large cloud of gas and dust in interstellar space. Some nebulae are regions where new stars are forming, while others are created by dying stars or stellar explosions. Radiation and gravity can shape nebulae into complex structures. Nebulae provide important information about the life cycle of stars.

26. Black Hole

A black hole is a region of spacetime where gravity is so strong that beyond its event horizon, nothing can escape to the outside. Black holes can form from the collapse of massive stars and may also grow through mergers and accretion. They can be detected indirectly through their effects on nearby matter, light, and spacetime.

27. Event Horizon

The event horizon is the boundary surrounding a black hole beyond which escape to the outside is impossible according to general relativity. It is not a solid surface. An observer far from the black hole cannot receive information from inside this boundary. The size of the event horizon depends on the black hole's mass and other properties.

28. Singularity

A singularity is a point or region in a mathematical description of spacetime where certain physical quantities become undefined or diverge. In classical general relativity, the center of an idealized black hole contains a singularity. Scientists expect a complete theory of quantum gravity may be needed to understand what actually occurs in such extreme conditions.

29. Supernova

A supernova is an extremely powerful stellar explosion that can occur when a star reaches the end of certain evolutionary paths. Some supernovae result from the collapse of massive stellar cores, while others involve thermonuclear destruction of white dwarfs. Supernovae release enormous energy and help distribute heavy elements into space.

30. Neutron Star

A neutron star is an extremely dense remnant left behind after the collapse and explosion of a massive star. It is composed primarily of matter packed at densities far greater than those found on Earth. Neutron stars can have very strong magnetic fields and may rotate rapidly. Some are observed as pulsars.

31. Pulsar

A pulsar is a rapidly rotating neutron star that emits beams of electromagnetic radiation from regions near its magnetic poles. As the star rotates, the beams can sweep across Earth and produce regular pulses of detected radiation. Pulsars are exceptionally precise natural clocks. They are used to study gravity, dense matter, and galactic environments.

32. Magnetar

A magnetar is a type of neutron star with an exceptionally strong magnetic field. Its magnetic energy can power intense bursts of X-rays and gamma rays. Magnetars are among the most extreme known objects in the universe. Their activity helps scientists study matter and magnetic fields under conditions impossible to reproduce on Earth.

33. White Dwarf

A white dwarf is the dense stellar remnant left after a low- or medium-mass star has exhausted the nuclear fuel available for its normal evolution. It is supported against gravitational collapse by electron degeneracy pressure. White dwarfs are roughly Earth-sized but can contain a substantial fraction of the Sun's mass. They slowly cool over time.

34. Red Giant

A red giant is a late stage in the evolution of a star in which the outer layers expand greatly after hydrogen fuel in the core becomes depleted. The expanded surface becomes cooler and therefore appears reddish. The core contracts while nuclear reactions continue in surrounding layers. Our Sun is expected to become a red giant in the distant future.

35. Blue Giant

A blue giant is a very massive, hot, luminous star with a surface temperature high enough to give it a blue or blue-white appearance. These stars consume nuclear fuel rapidly and therefore have relatively short lifetimes compared with smaller stars. Many end their lives in powerful supernova explosions. They enrich space with elements formed during stellar evolution.

36. White Hole

A white hole is a hypothetical solution to the equations of general relativity that behaves like a time-reversed black hole. Matter and energy could emerge from a white hole, but nothing could enter it from outside. No observational evidence has confirmed that white holes exist. They remain a topic of theoretical physics.

37. Wormhole

A wormhole is a hypothetical structure that would create a shortcut between distant regions of spacetime. It appears in certain mathematical solutions of general relativity, although there is no confirmed evidence that traversable wormholes exist. Stable wormholes may require unusual physical conditions. They are often studied in theoretical physics and discussions of spacetime geometry.

38. Dark Matter

Dark matter is a form of matter inferred from its gravitational effects but not directly observed through electromagnetic radiation. It appears to make up a large fraction of the matter in the universe. Evidence comes from galaxy rotation, gravitational lensing, and the growth of cosmic structures. Its exact physical nature remains unknown.

39. Dark Energy

Dark energy is the name given to the unknown component associated with the accelerated expansion of the universe. It appears to have a large effect on the universe at very large scales. Scientists do not yet know whether it is a property of space itself or a sign that current theories of gravity require modification.

40. Gravity

Gravity is the interaction that attracts objects with mass or, in general relativity, reflects the curvature of spacetime caused by mass and energy. It controls planetary orbits, stellar systems, galaxies, and large-scale cosmic structures. Gravity is also responsible for tides and the collapse of matter into stars.

and planets.

41. Light-Year

A light-year is a unit of distance equal to the distance light travels through vacuum in one year. It is used in astronomy because interstellar and intergalactic distances are extremely large. A light-year measures distance, not time. Because light takes time to travel, observing distant objects also means looking into the past.

42. Astronomical Unit

An astronomical unit, abbreviated AU, is a standard unit of distance based on the average Earth-Sun distance. It is approximately 150 million kilometers. Astronomers commonly use astronomical units when describing distances within planetary systems. The unit provides a convenient scale for comparing planetary orbits.

43. Parsec

A parsec is an astronomical unit of distance based on stellar parallax and is equal to about 3.26 light-years. It is widely used by astronomers when measuring distances to stars and galaxies. Larger units such as kiloparsecs and megaparsecs are used for galactic and cosmological distances.

44. Redshift

Redshift occurs when light from an astronomical object is shifted toward longer wavelengths. It can result from an object moving away from an observer, from the expansion of the universe, or from gravitational effects. Astronomers measure redshift to study motion, cosmic expansion, and the distances and histories of distant objects.

45. Blueshift

Blueshift occurs when light is shifted toward shorter wavelengths. It can indicate that an astronomical object is moving toward an observer relative to a chosen reference frame. Blueshift can also have other physical causes in specialized situations. Astronomers use spectral shifts to measure the motion of stars and galaxies.

46. Cosmic Microwave Background

The cosmic microwave background is faint electromagnetic radiation that fills the universe and is observed in the microwave region of the spectrum. It was released when the early universe cooled enough for atoms to form and light to travel freely. Its tiny temperature variations provide important information about the early universe and cosmic structure.

47. Big Bang

The Big Bang is the scientific model describing the early hot, dense state of the universe and its subsequent expansion and cooling. It does not describe an explosion occurring at one location in

pre-existing space. Instead, space itself has expanded over cosmic time. Evidence includes cosmic expansion, the cosmic microwave background, and the observed abundance of light elements.

48. Cosmology

Cosmology is the branch of science that studies the origin, large-scale structure, evolution, and future of the universe. It combines observations from astronomy with theories from physics and mathematics. Topics include cosmic expansion, dark matter, dark energy, the early universe, and the formation of galaxies and large-scale structures.

49. Exoplanet

An exoplanet is a planet located outside our Solar System that orbits another star or, in some cases, a stellar remnant. Thousands have been confirmed using methods such as transit observations and measurements of stellar motion. Exoplanets range from small rocky worlds to large gas giants. Their study helps scientists understand how planetary systems form.

50. Habitable Zone

The habitable zone is the region around a star where conditions could allow liquid water to exist on the surface of a suitable rocky planet. Its location depends on the star's brightness and other planetary conditions. Being in the habitable zone does not guarantee life. Atmosphere, geology, chemistry, and other factors also influence habitability.

51. Astrobiology

Astrobiology is an interdisciplinary field that investigates the origin, evolution, distribution, and potential future of life in the universe. It combines astronomy, biology, chemistry, geology, planetary science, and other disciplines. Researchers study Earth's extreme environments as well as Mars, icy moons, and exoplanets to understand where life might exist.

52. Extraterrestrial Life

Extraterrestrial life refers to life that originated outside Earth. No confirmed evidence of extraterrestrial life has yet been established. Scientists search for possible signs through observations of planets and moons, analysis of atmospheric gases, examination of space rocks, and exploration missions. The search includes both simple microbial life and more complex possibilities.

53. Space-Time

Spacetime is the combined mathematical description of the three dimensions of space and one dimension of time. In Einstein's theory of relativity, mass and energy can curve spacetime, and this curvature influences the motion of matter and light. The concept is fundamental to understanding gravity, black holes, and the large-scale universe.

54. Orbit

An orbit is the path followed by an object as it moves under the gravitational influence of another body or system. Planetary orbits are usually elliptical rather than perfectly circular. The shape and speed of an orbit depend on factors such as gravity, velocity, and the masses involved. Satellites and spacecraft are also placed into controlled orbits.

55. Elliptical Orbit

An elliptical orbit is an orbital path shaped like an ellipse rather than a perfect circle. Under gravity, many planets, moons, comets, and artificial spacecraft follow approximately elliptical paths. An orbit has a closest point and a farthest point from the central body. Orbital speed changes along an elliptical orbit.

56. Escape Velocity

Escape velocity is the minimum speed an object would need, in an idealized situation, to escape the gravitational influence of a body without additional propulsion. It depends on the mass and radius of the body. Earth has a much lower escape velocity than the Sun. Black holes have escape conditions so extreme that light cannot escape beyond the event horizon.

57. Orbital Period

An orbital period is the time required for an astronomical object to complete one orbit around another object. The period depends on the size of the orbit and the masses involved. Planets farther from the Sun generally have longer orbital periods. Artificial satellites also have orbital periods determined by their altitude and trajectory.

58. Solar Wind

The solar wind is a continuous flow of charged particles, mainly protons and electrons, released from the Sun's outer atmosphere. It travels through the Solar System and interacts with planetary magnetic fields and atmospheres. Solar wind can contribute to auroras and space-weather effects. Strong solar activity can make these interactions more intense.

59. Solar Flare

A solar flare is a sudden, intense release of electromagnetic energy from the Sun. Flares occur in active regions associated with strong magnetic fields and can produce radiation across many wavelengths. Powerful flares can affect radio communication and spacecraft systems near Earth. They are often associated with other forms of solar activity.

60. Coronal Mass Ejection

A coronal mass ejection, or CME, is a huge eruption of magnetized plasma from the Sun's corona. When directed toward Earth, a CME can interact strongly with Earth's magnetosphere and produce geomagnetic storms. Such storms may affect satellites, radio communication, navigation systems, and electrical infrastructure. CMEs are an important part of space weather.

61. Aurora

An aurora is a natural display of colored light in a planet's upper atmosphere caused by energetic charged particles interacting with atmospheric gases. On Earth, auroras are commonly seen near polar regions where magnetic field lines guide particles into the atmosphere. Auroras can appear green, red, blue, or violet depending on the gases and altitudes involved.

62. Magnetosphere

A magnetosphere is the region around a celestial body where its magnetic field strongly influences charged particles. Earth's magnetosphere helps deflect much of the solar wind and protects the atmosphere from some energetic particles. Other planets and moons may also have magnetospheres. Their shapes can change in response to solar activity.

63. Asteroid Belt

The asteroid belt is a region between the orbits of Mars and Jupiter containing millions of rocky bodies. Jupiter's strong gravitational influence prevented the material there from combining easily into a planet. Most asteroids are small, but the belt contains large bodies such as Ceres. It is a key region for studying early Solar System material.

64. Kuiper Belt

The Kuiper Belt is a distant region beyond Neptune containing many small icy bodies and dwarf planets. It is part of the Solar System's outer population of primitive objects. Pluto is one of its best-known members. Kuiper Belt objects preserve material from the early Solar System and help scientists study its formation.

65. Oort Cloud

The Oort Cloud is a hypothetical distant reservoir of icy objects believed to surround the Solar System at enormous distances. It has not been directly observed, but its existence is inferred from the orbital properties of some long-period comets. The Oort Cloud may extend far beyond the Kuiper Belt and mark the outermost region gravitationally associated with the Sun.

66. Spacecraft

A spacecraft is a vehicle or robotic machine designed to travel through outer space and perform scientific, communication, navigation, or exploration tasks. Spacecraft can be crewed or uncrewed. They use systems for power, communication, navigation, thermal control, and propulsion. Examples include orbiters, landers, rovers, telescopes, and probes.

67. Satellite

A satellite is an object that orbits another object in space. Natural satellites include moons, while artificial satellites are human-made spacecraft placed into orbit. Artificial satellites support communication, navigation, weather monitoring, Earth observation, and scientific research. Their orbits are selected

according to the mission they need to perform.

68. Space Probe

A space probe is an uncrewed spacecraft sent into space to collect scientific data. Probes can travel to planets, moons, asteroids, comets, or even beyond the Solar System. They carry instruments such as cameras, spectrometers, magnetometers, and particle detectors. Their observations greatly expand knowledge beyond what can be obtained from Earth.

69. Space Telescope

A space telescope is an astronomical observatory placed in space to observe objects and phenomena above Earth's atmosphere. Space telescopes can detect wavelengths that are absorbed or distorted by the atmosphere. They provide clearer observations of distant galaxies, stars, planets, and cosmic events. Different telescopes are designed for different parts of the electromagnetic spectrum.

70. Space Station

A space station is a large artificial spacecraft designed to support astronauts living and working in orbit for extended periods. It contains laboratories, living areas, life-support systems, and equipment for scientific experiments. Space stations allow researchers to study biology, physics, and materials under microgravity conditions.

71. Rocket

A rocket is a vehicle or propulsion system that produces thrust by expelling mass at high speed. Unlike aircraft engines, rockets can operate without atmospheric oxygen because they carry or generate the necessary propellant components. Rockets are used to launch spacecraft, satellites, and crewed vehicles. Different stages and engine designs are used for different missions.

72. Rocket Propulsion

Rocket propulsion is the process of generating thrust to accelerate a rocket by ejecting gases or other reaction mass. Chemical rockets obtain energy from combustion and are widely used for launch vehicles. Other propulsion methods include electric propulsion and proposed nuclear systems. Propulsion requirements depend on the mission, payload, and destination.

73. Microgravity

Microgravity refers to an environment in which objects experience very small apparent gravitational forces, such as inside an orbiting spacecraft. Astronauts and objects appear weightless because they are in continuous free fall around Earth. Microgravity affects human physiology, fluid behavior, combustion, and material processes. Scientists use it to conduct experiments that are difficult on Earth.

74. Zero Gravity

Zero gravity is a common informal expression used to describe a condition where apparent weight is extremely small. In orbit, gravity is still present, but astronauts experience weightlessness because they and their spacecraft are continuously falling together. Therefore, the term microgravity is often more scientifically accurate for space environments.

75. Space Radiation

Space radiation consists of energetic particles and electromagnetic radiation found beyond Earth's protective atmosphere and magnetic field. Sources include the Sun, distant stars, and cosmic-ray-producing events. High radiation exposure can harm biological tissue and spacecraft electronics. Space missions therefore require shielding, monitoring, and careful mission planning.

76. Cosmic Rays

Cosmic rays are highly energetic particles traveling through space, many of which originate outside the Solar System. They are primarily atomic nuclei and protons moving at extremely high speeds. When cosmic rays enter Earth's atmosphere, they produce showers of secondary particles. Their study provides information about energetic processes in the universe.

77. Gravitational Wave

A gravitational wave is a traveling disturbance in spacetime produced by accelerating massive objects under appropriate conditions. Powerful sources include merging black holes and neutron stars. These waves change distances by extremely small amounts as they pass through Earth. Detectors such as LIGO and Virgo have allowed scientists to observe these events directly.

78. Gravitational Lensing

Gravitational lensing occurs when the gravity of a massive object bends the path of light from a more distant object. This can magnify, distort, or produce multiple images of the background source. Astronomers use gravitational lensing to study distant galaxies and to map otherwise invisible matter such as dark matter.

79. Quasar

A quasar is an extremely luminous active galactic nucleus powered by matter falling toward a supermassive black hole. As material forms a hot accretion disk, enormous amounts of energy can be released. Quasars can be visible across immense distances and are among the brightest persistent objects in the universe. Their light provides information about the early universe.

80. Accretion Disk

An accretion disk is a rotating disk of gas, dust, or plasma that forms around a massive object as material falls inward under gravity. Friction and other processes heat the material, causing it to emit radiation. Accretion disks occur around young stars, white dwarfs, neutron stars, and black holes. They are important sources of high-energy radiation.

81. Supermassive Black Hole

A supermassive black hole is a black hole with a mass ranging from millions to billions of times the mass of the Sun. Such objects are found at the centers of many large galaxies, including the Milky Way. Their growth is connected to galaxy evolution. When actively feeding, they can power extremely bright galactic centers.

82. Stellar Evolution

Stellar evolution is the series of physical changes a star undergoes during its lifetime. A star's mass strongly influences its temperature, luminosity, nuclear reactions, and eventual fate. Stars can progress through stages such as main sequence, giant phases, and compact remnants. Studying stellar evolution explains how stars produce and distribute elements.

83. Main Sequence

The main sequence is the stage of stellar evolution during which a star steadily fuses hydrogen into helium in its core. Most stars spend the majority of their lifetimes in this stage. Their position on the main sequence depends strongly on mass. More massive stars are generally hotter and brighter but consume their fuel more rapidly.

84. Nuclear Fusion

Nuclear fusion is a process in which light atomic nuclei combine to form heavier nuclei while releasing energy under suitable conditions. Fusion powers the Sun and other stars. Extremely high temperatures and pressures are needed to overcome the electrical repulsion between nuclei. Scientists are also researching controlled fusion as a potential energy technology.

85. Nuclear Fission

Nuclear fission is a nuclear process in which a heavy atomic nucleus splits into smaller nuclei and releases energy, radiation, and often additional neutrons. Fission can occur spontaneously or be induced by particle interactions. It is used in nuclear reactors and weapons. The process can sustain a chain reaction under appropriate conditions.

86. Planetary System

A planetary system is a star or stellar remnant together with planets, moons, asteroids, comets, dust, and other objects gravitationally bound to it. Planetary systems form from disks of gas and dust around young stars. Their architectures can vary greatly. Studying them helps scientists understand how planets form and evolve.

87. Protoplanetary Disk

A protoplanetary disk is a rotating disk of gas and dust surrounding a young star. Material within the disk can collide and gradually form planetesimals and planets. These disks also contain the raw materials from which moons, asteroids, and other bodies can develop. Astronomers observe them around many

young stars.

88. Planet Formation

Planet formation is the process by which planets develop from material surrounding a young star. Dust grains collide and stick together, forming larger bodies called planetesimals, which can eventually grow into planetary embryos and planets. Gravity, collisions, migration, and the surrounding disk environment all influence the final planetary system.

89. Tidal Force

Tidal force is a differential gravitational force that occurs because gravity is stronger on the side of an object closer to another massive body. On Earth, the Moon and Sun produce tidal effects that contribute to ocean tides. Tidal forces can also heat moons and influence their rotation and orbital evolution.

90. Tidal Locking

Tidal locking occurs when an orbiting body's rotation period becomes synchronized with its orbital period around another body. As a result, the same side of the smaller body continually faces the larger one. Earth's Moon is tidally locked to Earth. Tidal interactions over long periods can gradually change rotation and orbital properties.

91. Planetary Rings

Planetary rings are collections of dust, ice, rock, or other small particles orbiting a planet. Saturn has the most prominent ring system, but Jupiter, Uranus, and Neptune also have rings. Rings can form from broken moons, leftover material, or other processes. Their structures are shaped by gravity and interactions with nearby moons.

92. Atmosphere

An atmosphere is a layer of gases held around a celestial body by gravity. Planetary atmospheres can differ greatly in composition, pressure, temperature, and density. They influence surface conditions, weather, climate, and the interaction between a planet and space. Earth has a nitrogen- and oxygen-rich atmosphere, while other planets have very different compositions.

93. Terraforming

Terraforming is a hypothetical process of deliberately modifying a planetary environment to make it more suitable for Earth-like life. Proposed changes might involve atmosphere, temperature, water availability, or surface chemistry. No complete terraforming project has been carried out. The idea raises major scientific, engineering, ethical, and environmental questions.

94. Space Weather

Space weather describes changing conditions in space caused mainly by solar activity and their effects on planets, spacecraft, and technology. Solar flares, coronal mass ejections, and the solar wind can

influence Earth's magnetosphere and upper atmosphere. Strong space-weather events can affect satellites, communication, navigation, and power systems.

95. Planetary Geology

Planetary geology is the study of the physical structure, composition, and geological processes of planets, moons, asteroids, and other solid bodies. Scientists examine craters, volcanoes, mountains, tectonic features, minerals, and surface materials. Comparing different worlds helps reveal how planets form and evolve.

96. Astrochemistry

Astrochemistry is the study of the chemical composition and reactions of matter in space. It examines molecules in interstellar clouds, planetary atmospheres, comets, stars, and protoplanetary disks. Astrochemistry helps explain how complex molecules form and how the chemical ingredients needed for planets and life are distributed.

97. Radio Astronomy

Radio astronomy is the study of astronomical objects using radio waves rather than visible light. Radio telescopes can detect emissions from gas clouds, pulsars, galaxies, and other sources. Radio observations are especially useful for studying cold gas and energetic processes. They also allow astronomers to observe through some dust that blocks visible light.

98. Spectroscopy

Spectroscopy is a scientific technique used to study how matter interacts with electromagnetic radiation. By examining the wavelengths absorbed or emitted by an object, scientists can determine information about its composition, temperature, motion, and physical conditions. Astronomers use spectroscopy extensively to analyze stars, planets, galaxies, and nebulae.

99. Astronomy

Astronomy is the scientific study of objects and phenomena beyond Earth's atmosphere. It includes the study of stars, planets, galaxies, black holes, cosmic radiation, and the evolution of the universe. Astronomers use observations, mathematics, physics, computer modeling, and spacecraft data to investigate the cosmos.

100. Astrophysics

Astrophysics is the branch of astronomy that applies the laws of physics to understand astronomical objects and cosmic phenomena. It investigates topics such as stellar interiors, black holes, galaxies, radiation, and the origin of cosmic structures. Astrophysicists use theoretical models together with observations to explain how the universe behaves.